Almost sure convergence of supercritical Liouville first passage percolation

Research manuscript

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Abstract

We prove almost sure convergence of heat-kernel Liouville first passage percolation in the supercritical phase, in the topology on lower semicontinuous metrics and with the usual median crossing normalization. The main estimate compares distances between shrinking endpoint neighborhoods with a multiplicative constant arbitrarily close to one and polynomially small failure probability. To improve the comparison constant, we use separating paths whose lengths are measured in the other metric, together with a covering argument in the time parameter of a geodesic. The estimate is uniform over geodesics contained in a fixed Euclidean ball. Rational-circle distances identify the constant and provide the normalizer control needed to pass from a polynomial mesh to all positive real cutoffs.

1 Introduction

Let h be a whole-plane Gaussian free field with the additive constant fixed so that $h_1(0) = 0$, where $h_r(z)$ is its circle average over $\partial B_r(z)$. Fix $\xi > \xi_{\text{crit}}$. Set

$$h^*_\varepsilon = h * p_{\varepsilon^2/2}, \qquad p_t(z) = \frac{1}{2\pi t} e^{-|z|^2/(2t)}, \qquad D^\varepsilon_h(z,w) = \inf_{P:z\to w} \int_0^1 e^{\xi h^*_\varepsilon(P(t))} |P'(t)|\, dt. \tag{1.1}$$

The infimum is over piecewise continuously differentiable paths. Let $\mathfrak{a}_\varepsilon$ be the median of the left-right crossing distance of $[0,1]^2$, with paths restricted to the square, and write $X_\varepsilon = \mathfrak{a}_\varepsilon^{-1} D^\varepsilon_h$. Let $D = D_h$ be the limit in probability constructed in [1, Theorems 1.3 and 1.8], with this same normalization. Write $Q = Q(\xi)$ for the exponent determined by $\mathfrak{a}_\varepsilon = \varepsilon^{1-\xi Q+o(1)}$.

Theorem 1.1. *For each fixed $\xi > \xi_{\text{crit}}$, almost surely,*

$$\mathfrak{a}_\varepsilon^{-1} D^\varepsilon_h \longrightarrow D_h \quad \text{as } \varepsilon \to 0$$

in the topology on lower semicontinuous functions on $\mathbb{C}\times\mathbb{C}$. The limit is through all positive real values of ε, on the probability space carrying the original field h.

Precisely, the topology in Theorem 1.1 requires the following two properties. If $\varepsilon_n \to 0$ and $(z_n, w_n) \to (z, w)$, then

$$D(z,w) \le \liminf_{n\to\infty} X_{\varepsilon_n}(z_n, w_n). \tag{1.2}$$

For each (z,w) and each sequence $\varepsilon_n \to 0$, there must also be $(z_n, w_n) \to (z,w)$ for which the corresponding distances converge to $D(z,w)$. These statements are to hold on one probability-one event. Distances equal to ∞ are allowed. No assertion simultaneously for every value of ξ is made.

The difficult step is the improvement of the fixed comparison constant in [2, Proposition 3.6]. The proof below is organized around this step. We first localize convergence in probability without assuming convergence of internal metrics. We then construct short separating paths in one metric near geodesics of the other metric. A deterministic interval-covering argument gives a contraction of the comparison constant. Finally, we use rational-circle distances to identify the constant and to control the normalizers between successive cutoffs of a polynomial mesh.

1.1 Notation and imported results

Write $B_r(z)$ for an open Euclidean ball and $\mathbb{A}_{r,R}(z) = B_R(z) \setminus \overline{B}_r(z)$. A path *around* an annulus is a continuous path in the annulus whose image disconnects its boundary components. It need not be simple. The distance around an annulus is the infimum of the lengths of such paths. Distances across annuli have the usual crossing meaning. For a closed set V, $d(u, v; V)$ is the infimum of the d-lengths of continuous paths contained in V with endpoints u, v. For open sets we use the same convention.

Throughout, "with polynomially high probability" means with probability at least $1 - O(\varepsilon^\beta)$ for some $\beta > 0$. The exponent and the implicit constant may depend on fixed parameters. They need not be uniform as those parameters approach a boundary of their permitted range.

We use the following results, all valid for the fixed supercritical parameter considered here.

(i) $X_\varepsilon \to D_h$ in probability in the lower semicontinuous topology, also with h replaced by $h + f$ for a bounded continuous f. The metric is a measurable function of the field, is local, satisfies Weyl scaling, and for deterministic $r > 0, z \in \mathbb{C}$ satisfies

$$D_h(z + r\cdot, z + r\cdot) = r^{\xi Q} e^{\xi h_r(z)} D_{h^{z,r}}, \qquad h^{z,r} = h(z + r\cdot) - h_r(z). \tag{1.3}$$

See [1, Theorems 1.3 and 1.8, Definition 1.7].

(ii) Finite-distance pairs admit D-geodesics. Finite-length paths are Euclidean continuous; on Euclidean compact sets the Euclidean metric is bounded by a random constant times a positive power of D. Furthermore,

$$D(\overline{B}_k(0), \partial B_R(0)) \longrightarrow \infty \quad \text{as } R \to \infty \tag{1.4}$$

almost surely, for each k. See [4, Propositions 1.10 and 1.12, Lemma 3.12].

(iii) For two fixed disjoint rational circles O, O', $0 < D(O, O') < \infty$ almost surely and

$$X_\varepsilon(O, O') \longrightarrow D(O, O') \quad \text{in probability.} \tag{1.5}$$

Indeed, [4, Theorem 1.7] gives almost sure convergence of these distances along an extracted subsequence of every sequence. Its metric limit agrees with D by (i), so the subsequence criterion gives (1.5).

(iv) We use the localized LFPP comparison and scaling-ratio estimates in [2, Proposition 3.6, Lemma 3.7 and Theorem 1.9]. These are stated explicitly below. Tightness of annular crossing distances and their reciprocals, and of annular separating-path lengths, follows from [2, Lemma 3.1 and its proof]. For LFPP in an arbitrary fixed annulus, one can use [6, Proposition 4.1]: lower bounds follow from distances between its boundary components in a larger domain, and upper bounds around the annulus follow by joining finitely many fixed points using internal paths in angular sectors. The resulting concatenation has winding number one and its normalized length is bounded by a finite sum of tight random variables.

(v) We use the following annulus concentration estimate [3, Lemma 2.12]. If the events E_j are measurable from the field in sufficiently separated concentric annuli, modulo additive constants, then, given $a > 0$ and $b \in (0,1)$, there is $p < 1$ such that

$$\mathbb{P}[E_j] \geq p \text{ for every } j \quad \Longrightarrow \quad \mathbb{P}[\#\{j \leq n : E_j\} < bn] \leq Ce^{-an}. \tag{1.6}$$

The separation ratio is fixed before p is chosen. In particular, the rate a can be made arbitrarily large by raising the required marginal probability.

No estimate which bounds $D(u,v)$ uniformly over all sufficiently close u, v is used. Such an estimate would be false in the present setting.

2 Localization and preliminary consequences

2.1 The localized field

Choose a smooth radial cutoff $\chi : [0,\infty) \to [0,1]$ which is one on $[0,1/2]$ and zero on $[1,\infty)$. For $0 < \varepsilon < e^{-1}$, let $\widehat{h}_\varepsilon$ be the convolution obtained by multiplying $p_{\varepsilon^2/2}$ by $\chi(|\cdot|/(\varepsilon \log \varepsilon^{-1}))$ and dividing by its integral. Write $\widehat{D}^\varepsilon_h$ for its path metric and $\widehat{X}_\varepsilon = \mathfrak{a}_\varepsilon^{-1} \widehat{D}^\varepsilon_h$. Its internal metric on U is measurable from h on $B_{\varepsilon \log \varepsilon^{-1}}(U)$.

Lemma 2.1. *Almost surely, for every compact K,*

$$\sup_K |\widehat{h}_\varepsilon - h^*_\varepsilon| \longrightarrow 0. \tag{2.1}$$

For every $a > 0$, almost surely,

$$\sup_{\varepsilon,\delta \in [(n+1)^{-a}, n^{-a}]} \sup_K |h^*_\varepsilon - h^*_\delta| \longrightarrow 0. \tag{2.2}$$

The following uniform version of (2.1) also holds. If $H_{\delta,L}$ uses the cutoff radius δL, then, for $L \geq \log \delta^{-1}$,

$$\sup_K |H_{\delta,L} - h^*_\delta| \leq C_K (1 + \log \delta^{-1})^3 e^{-(\log \delta^{-1})^2/8} \tag{2.3}$$

for all sufficiently small δ, with a finite random C_K.

Proof. The circle-average bound [2, Lemma 2.3], with its parameter $\zeta = 1/2$, gives $|h_s(z)| \leq C_K(1 + |\log s|)$ for $z \in K$, $s > 0$. Radial integration expresses a heat-kernel convolution as

$$h^*_\delta(z) = \int_0^\infty \frac{2s}{\delta^2} e^{-s^2/\delta^2} h_s(z)\, ds.$$

The omitted mass for $H_{\delta,L}$ is at most $e^{-L^2/4}$. Integrating the circle-average bound over $s \geq \delta L/2$, and including the normalization error, gives a constant times $(1+\log \delta^{-1}+L)^3 e^{-L^2/4}$. Its supremum over $L \geq \log \delta^{-1}$ is bounded by the right side of (2.3). This proves (2.1) as well.

Differentiating the radial kernel with respect to $\log \delta$, the same bound gives $\sup_K |\partial_{\log \delta} h^*_\delta| \leq C_K(1 + \log \delta^{-1})$ for small δ. The logarithmic width of $[(n+1)^{-a}, n^{-a}]$ is $a \log(1 + 1/n)$. Their product tends to zero, proving (2.2). This is also the field-only content of [5, Lemma 4.2]; no subcritical metric estimate is involved. □

Under the change of coordinates $w = z + ru$, the localized field becomes $H_{\delta,L}$ for $h^{z,r}$, up to $h_r(z)$, where

$$\delta = \varepsilon/r, \qquad L = \log \varepsilon^{-1}. \tag{2.4}$$

For $r \le 1$ we have $L \ge \log \delta^{-1}$. For $r \ge \varepsilon^t$ with fixed $t < 1$, $\delta L \to 0$ uniformly. Thus we may use fixed-scale statements uniformly for the dilated localized metrics. Define the deterministic ratio

$$\kappa_\varepsilon(r) = \frac{r\mathfrak{a}_{\varepsilon/r}}{r^{\xi Q}\mathfrak{a}_\varepsilon}. \tag{2.5}$$

By [2, Lemma 3.7 and Theorem 1.9], for each $\zeta \in (0,1)$ there is C_ζ such that

$$C_\zeta^{-1} \le \kappa_\varepsilon(r) \le C_\zeta, \qquad 100\varepsilon^{1-\zeta} \le r \le 1, \tag{2.6}$$

for all sufficiently small ε. The constant in (2.6) does not depend on r or ε.

2.2 Fixed-constant comparison and confinement

We record [2, Proposition 3.6] in the form we need: for each $\zeta \in (0,1)$ there is C_0 such that for every deterministic bounded connected open U, with polynomially high probability,

$$\widehat{X}_\varepsilon(B_\rho(z), B_\rho(w); B_\rho(U)) \le C_0 D(z,w;U), \tag{2.7}$$

$$D(B_\rho(z), B_\rho(w); B_\rho(U)) \le C_0 \widehat{X}_\varepsilon(z,w;U), \tag{2.8}$$

simultaneously for $z, w \in U$, where $\rho = \varepsilon^{1-\zeta}$. The constant C_0 depends on ζ and the metric law but not on U. Our cutoff is exactly an allowed choice in [2, Section 2.3], with its parameter $q = 1$.

Lemma 2.2. *For each compact K and $M < \infty$, the probability that an $\widehat{X}_\varepsilon$-path of length at most M starting in K exits $B_R(0)$ tends to zero as first $\varepsilon \to 0$ and then $R \to \infty$. The same holds for X_ε. Moreover, for two fixed disjoint rational circles,*

$$\widehat{X}_\varepsilon(O, O') \longrightarrow D(O, O') \quad \textit{in probability.} \tag{2.9}$$

Proof. Take $U = B_{R+1}(0)$ in (2.8). Stop a path on its first hit of $\partial B_R(0)$. If such a path has length at most M, then

$$D(B_\rho(K), B_\rho(\partial B_R(0))) \le C_0 M.$$

By lower semicontinuity and compactness, the left side has lower limit at least $D(K, \partial B_R(0))$. Now use (1.4), with K contained in a fixed disk. The argument for ordinary LFPP follows from (2.1) on $B_{R+1}(0)$.

The ordinary circle distance is tight by (1.5). Its nearly minimizing paths are confined to a fixed compact set with probability arbitrarily close to one. Comparing their lengths by (2.1) gives tightness of the localized circle distance and one inequality between the two distances, up to a factor tending to one in probability. The confinement just proved for localized LFPP gives the reverse inequality. Hence their ratio tends to one in probability, and (2.9) follows. On these confinement events the minimizing paths can be taken to be actual geodesics: paths below a fixed length threshold are confined to a compact set, on which the smooth positive Riemannian density gives existence by the usual compactness argument. □

3 Local comparisons and separating paths

3.1 A consequence of lower semicontinuous convergence

For a function f on a Euclidean space put $f^{[\eta]}(p) = \inf_{|q-p|<\eta} f(q)$; for pairs of points use the maximum product distance.

Lemma 3.1. *Suppose that nonnegative lower semicontinuous functions f_n converge to f in the lower semicontinuous topology. For each compact K, $M < \infty$, and $\eta, \nu > 0$, eventually*

$$f_n^{[\eta]}(p) \le f(p) + \nu \qquad \text{if } p \in K,\ f(p) \le M,$$
$$f^{[\eta]}(p) \le f_n(p) + \nu \qquad \text{if } p \in K,\ f_n(p) \le M.$$

The corresponding events have probability tending to one if convergence holds in probability.

Proof. The truncated epigraph $\{(p,t) : p \in K,\ f(p) \le t \le M\}$ is compact. Cover it by finitely many product neighborhoods of spatial radius less than $\eta/3$ and height radius less than $\nu/3$, centered at (p_i, t_i) in this epigraph. Take a recovery sequence at each p_i. Eventually its points lie within $\eta/3$ of p_i and their values are at most $f(p_i) + \nu/3 \le t_i + \nu/3$. Applying the cover at $(p, f(p))$ proves the first inequality.

If the second fails along a subsequence, take $p_n \in K$ with $f_n(p_n) \le M$ and $f^{[\eta]}(p_n) > f_n(p_n) + \nu$. Extract a further subsequence on which $p_n \to p$ and $f_n(p_n) \to t \le M$. The lower-limit property gives $f(p) \le t$. For large n, p is in the η-ball of p_n, a contradiction. The in-probability statement follows by extracting an almost surely convergent subsequence from any sequence along which the failure probability were bounded away from zero. □

Lemma 3.2. *Let*

$$V = \{1/2 \le |z| \le 3\}, \qquad U = \{1/8 \le |z| \le 16\}.$$

For each $M < \infty$ and $\eta, \nu > 0$, with probability tending to one as $\delta \to 0$, simultaneously for $|u| = 1, |v| = 2$,

$$X_\delta(B_\eta(u), B_\eta(v); U) \le D(u, v; V) + \nu \qquad \text{if } D(u,v;V) \le M, \tag{3.1}$$
$$D(B_\eta(u), B_\eta(v); U) \le X_\delta(u, v; V) + \nu \qquad \text{if } X_\delta(u,v;V) \le M. \tag{3.2}$$

The same holds for X_δ constructed from $H_{\delta,L}$, uniformly over $L \ge \log \delta^{-1}$. If $\delta L \to 0$, the latter event is measurable from the field in $\mathbb{A}_{1/16,32}(0)$ modulo its additive constant, when all lengths and additive errors are expressed in units $e^{\xi h_1(0)}$.

Proof. We may decrease η so that $\eta < 1/10$. Fix a smooth nonnegative function f which is zero on a neighborhood of V and equals a constant B on two closed annular collars separating V from the two boundary components of U. Use narrower collars inside the constant regions to accommodate mollification. Positivity of D-crossings and tightness away from zero of normalized LFPP crossings imply that, by taking B large, the probability that every $(h+f)$-path from $B_\eta(V)$ to the complement of U costs more than $M + 3$ can be made arbitrarily close to one, in both metrics, uniformly for small δ.

Apply Lemma 3.1 to the convergence in probability for $h + f$, with height cutoff $M + 1$, on the compact product of the two endpoint circles. A D-path in V has unchanged length after adding f. An LFPP path in V changes its length by a factor $e^{o(1)}$, since $\sup_V |f_\delta^*| \to 0$. Choose the error in Lemma 3.1 smaller than $\nu/4$. The resulting target paths of length at most $M + 2$ are confined to U by the collars. Removing the nonnegative f decreases their lengths. This proves both inequalities,

with arbitrarily high limiting probability. Since B was only used to prove the probability bound and does not occur in the asserted event, its probability tends to one.

Lemma 2.1 transfers this conclusion to $H_{\delta,L}$ uniformly in L, by first using cutoff $2M$ and a smaller additive error. Both source and target paths in the resulting event stay in fixed closed annuli. Locality of D and the support of the localized convolution give the final measurability statement. The additive error rescales together with the two metrics under addition of a field constant. □

Remark 3.3. The universal endpoint statements above can be treated as measurable events, without a separability assumption in the Euclidean topology for the metric itself. Here are details. On a fixed closed annulus, the internal metrics are lower semicontinuous: parameterize paths of bounded length at constant speed, use the local inverse Hölder bound to obtain Euclidean equicontinuity, and pass to a subsequence. Lower semicontinuity bounds the length of the limit by the lower limit of the lengths. More precisely, for a length bound T, use the compact family of paths on $[0,1]$ satisfying $d(P(s),P(t)) \le T|s-t|$. These conditions are closed by lower semicontinuity and can be checked at rational times. Every path of length at most T has such a parameterization. Evaluation of path length is the supremum over rational partitions, hence is jointly measurable. Projection over the compact path families gives a measurable random internal lower semicontinuous metric; one may localize on countably many bounds for the inverse Hölder constants. For smooth LFPP the same argument uses bounds on its positive density on the annulus. Locality on a closed annulus follows by computing path lengths with the internal metric on a slightly larger open annulus: its boundary is at positive D-distance from the smaller annulus, so sufficiently short pieces of any rectifiable path have the same internal and ambient distance.

For any lower semicontinuous g, $g^{[\eta]}$ is upper semicontinuous: any point witnessing a strict upper bound continues to witness it after a sufficiently small change of the center. The inequalities of Lemma 3.2 are therefore measurable maximum conditions on a compact truncated epigraph of the source internal metric. Alternatively, their failure sets are projections of jointly Borel sets with compact sections after specifying a positive rational strict gap. These observations also justify the probabilistic use of Lemma 3.1. Locality can be taken with respect to the completed field sigma algebra.

3.2 Short paths measured in the other metric

Lemma 3.4. *Let d, d' be planar length metrics with Euclidean-continuous rectifiable paths. Let $r_1 > 4r_2 > \cdots > 4^{n-1}r_n$ and put*

$$Y_i = d(\text{across}\mathbb{A}_{r_i,2r_i}(c)), \quad W_i = d(\text{around}\mathbb{A}_{2r_i,3r_i}(c)), \quad Z_i = d'(\text{around}\mathbb{A}_{2r_i,3r_i}(c)).$$

Suppose $0 < Y_i < \infty$, $W_i \le CY_i$, and $Z_i \le CY_i$. If a d-geodesic segment I has endpoints outside $B_{3r_1}(c)$ and visits $B_{r_n}(c)$, then

$$Z_n \le \frac{C}{2}(1+2/C)^{-(n-1)}\,\mathrm{len}(I;d). \tag{3.3}$$

Proof. A separating path in the ith outer annulus is hit by I before and after its visit to the innermost disk. The intervening geodesic segment crosses each inner crossing annulus twice. It can be replaced by the portion of the separating path between these two intersection points. Taking infima gives

$$2\sum_{j>i} Y_j \le W_i \le CY_i.$$

Writing $S_i = \sum_{j \geq i} Y_j$, we get $S_i \geq (1 + 2/C)S_{i+1}$. Also $2S_1 \leq \mathrm{len}(I; d)$ by the disjoint crossings. Thus $Y_n \leq (1 + 2/C)^{-(n-1)} \,\mathrm{len}(I; d)/2$, proving the claim. □

Write $Y_{\delta,L}$ for the LFPP metric built from $H_{\delta,L}$ and divided by $\mathfrak{a}_\delta$. This notation keeps track of the localization window after dilation.

Lemma 3.5. *Fix a compact $K \subset \mathbb{A}_{3/4,5/2}(0)$ and $q > 0$. There are $\theta > 0, C < \infty$ such that for every sufficiently small $\eta > 0$ there are $\delta_0, \ell_0 > 0$ with the following property. If $0 < \delta < \delta_0$, $L \geq \log \delta^{-1}$, and $\delta L < \ell_0$, the following holds on an event of probability at least $1 - C\eta^q$. Simultaneously for $(d, d') = (D, Y_{\delta,L})$ and $(Y_{\delta,L}, D)$, for every $p \in K$ on a d-geodesic segment I whose endpoints lie outside $B_{2\eta^{1/2}}(p)$, there is a d'-rectifiable separating path whose image disconnects $B_\eta(p)$ from infinity, is contained in $B_{2\eta^{1/2}}(p)$, and has length at most*

$$C\eta^\theta \,\mathrm{len}(I; d). \tag{3.4}$$

The event is measurable from the field in $\mathbb{A}_{1/2,3}(0)$ modulo additive constants. No locality assertion is made here for the untruncated heat-kernel metric.

Proof. Use a lattice of mesh $\eta/10$ covering K, and around each lattice point take concentric annuli at a fixed sufficiently large geometric separation, with crossing radii between 4η and $\eta^{1/2}/4$. For every such radius s, annular tightness and (2.6), used with $\zeta = 1/2$ and δ small enough that $s \geq 100\delta^{1/2}$, give the required upper and lower bounds for the ordinary LFPP annular quantities. The rescaled cutoff is $H_{\delta/s,L}$; since $L \geq \log(s/\delta)$, (2.3) transfers these bounds to $Y_{\delta,L}$ with uniformly high marginal probability. Thus the hypotheses of Lemma 3.4 hold in both directions with a single constant C_1 and marginal probability as close to one as desired. This uses only tightness, not convergence of annular distances. The constant C_1 is independent of η, δ, L.

The dependence neighborhoods of the localized field fit inside slightly enlarged, still separated annuli once $\delta L \ll \eta$. Apply (1.6), choosing its exponential rate large enough to absorb a union bound over the $O(\eta^{-2})$ lattice points. Except on an event of probability $O(\eta^q)$, each point has at least $c \log \eta^{-1}$ successful annuli. Lemma 3.4 then gives (3.4), including a factor of two for a nearly minimizing separating path. If p is replaced by a lattice point within $\eta/10$, the innermost disk still contains $B_\eta(p)$ and the outermost annulus remains inside $B_{2\eta^{1/2}}(p)$. The segment I enters the innermost disk and its endpoints remain outside the outermost annulus. This proves the simultaneous assertion.

All events just used are finite annular comparisons. Their supports lie inside $\mathbb{A}_{1/2,3}(0)$ when η is small. The uniform cutoff statement follows from (2.3) and the same annular tightness argument after rescaling. Ratios of annular distances are invariant under addition of a constant to the field. □

3.3 Good root scales

Choose a probability $p_0 < 1$, sufficiently close to one for the application of (1.6) in Section 4. This choice is fixed for the rest of the argument. By annular tightness, choose $A \geq 1$, $m > 0$ and $M < \infty$ such that with probability at least $1 - (1 - p_0)/4$, in each of the two canonically normalized metrics,

$$d(\mathrm{around}\mathbb{A}_{7,8}) \leq A\, d(\mathrm{across}\mathbb{A}_{1,2}), \quad d(\mathrm{across}\mathbb{A}_{1,2}) \geq m, \quad d(\mathrm{around}\mathbb{A}_{7,8}) \leq M. \tag{3.5}$$

Here the two canonically normalized metrics are D and $Y_{\delta,L}$, and all annuli without a specified center are centered at zero. Uniform tightness for $Y_{\delta,L}$ follows from Lemma 2.1. Fix $d_0 > 0$. Choose $\eta > 0$ so small that Lemma 3.5 fails with probability at most $(1 - p_0)/4$ and its bound is at

most $(d_0/10)\,\mathrm{len}(I;d)$, and also $2\eta^{1/2} < 1/10$. The localized version of Lemma 3.2, with endpoint radius $\eta/2$ and additive error smaller than $d_0 m/20$, fails with probability at most $(1-p_0)/4$ for all sufficiently small cutoffs and dependence radii.

Intersect these events. After allowing smaller preliminary errors for nearly minimizing paths, the resulting event has probability at least p_0 and the following consequences in each direction: a source geodesic crossing segment in $\overline{\mathbb{A}}_{1,2}$ of length at most M has a target path between its $\eta/2$ endpoint balls of length at most $(1+d_0/10)$ times its source length; and the endpoints of such a segment can be surrounded by the paths from Lemma 3.5. The source internal metric in Lemma 3.2 is evaluated on V, so there is no boundary ambiguity.

At center z and physical radius r, define the event $E_{\varepsilon,r}(z)$ by this construction in coordinates $z + r\cdot$. Divide D by $r^{\xi Q} e^{\xi h_r(z)}$ and divide $\widehat{X}_\varepsilon$ by $r\mathfrak{a}_{\varepsilon/r}\mathfrak{a}_\varepsilon^{-1} e^{\xi h_r(z)}$. These are the two canonical units. Consequently target costs in physical units acquire the factor $\kappa_\varepsilon(r)$ in the D-source direction and $\kappa_\varepsilon(r)^{-1}$ in the other direction. The event is measurable from h in $\mathbb{A}_{r/16,32r}(z)$ modulo constants. In every fixed exponent band $\varepsilon^t \le r \le \varepsilon^s$, $0 < s < t < 1$, its marginal probability is at least p_0 for all sufficiently small ε, uniformly in r, z.

The constants A, m, M were chosen before d_0. In particular,

$$\lambda = \frac{1}{5A} > 0 \tag{3.6}$$

is independent of d_0 and of every exponent band. This independence is needed for the iteration.

4 Contraction of the comparison constant

4.1 A deterministic covering and joining argument

Lemma 4.1. *Let $P : [0, L] \to B_R(0)$ be a source geodesic parameterized by source length, with endpoints z, w. Suppose that for each point of P there is a center c within $r_{\min}/10$ and a good radius $r \in [r_{\min}, r_{\max}]$. Suppose a prior comparison with constant C_0 and endpoint radius $\rho_{\mathrm{old}} < \eta r_{\min}/4$ applies to every subsegment of P. Let K bound the target scaling factor at all the chosen good radii. Then*

$$d'(B_{513r_{\max}}(z), B_{513r_{\max}}(w)) \le F(C_0, K)L, \quad F(C,K) = (1-\lambda)\max\{C,K\} + (\lambda + 3d_0)K. \tag{4.1}$$

Proof. The conclusion is immediate if $|z-w| \le 1024 r_{\max}$, since the two open output balls, each of radius $513 r_{\max}$, overlap.

Otherwise, let b_0 be the first hit of $\overline{B}_{64r_{\max}}(w)$ and let a_0 be the last visit to $\overline{B}_{64r_{\max}}(z)$ before b_0. Put $J = [a_0, b_0]$. For each $\tau \in J$ choose a good disk $B_r(c)$ containing $P(\tau)$ in $B_{r/2}(c)$. The global endpoints lie outside $B_{32r}(c)$. Let I_τ be the time interval from the last visit to $\partial B_{6r}(c)$ before τ to the first such visit after τ. Its interior contains τ, and its image is in $\overline{B}_{6r}(c)$. By using a source separating path in $\mathbb{A}_{7r,8r}(c)$ as a shortcut, as in Lemma 3.4,

$$|I_\tau| \le d(\mathrm{around}\mathbb{A}_{7r,8r}(c)).$$

Let t_τ be the first hit after τ of $\partial B_{2r}(c)$, and let s_τ be the last hit of $\partial B_r(c)$ before t_τ. Then $G_\tau = [s_\tau, t_\tau] \subset I_\tau$, its image lies in $\overline{\mathbb{A}}_{r,2r}(c)$, and

$$|G_\tau| \ge d(\mathrm{across}\mathbb{A}_{r,2r}(c)) \ge |I_\tau|/A. \tag{4.2}$$

This also shows that the crossing segment has normalized source length at least m, whereas its length is at most the normalized bound M for $|I_\tau|$.

Take a finite subcover of J by the interiors of I_τ. Greedily select disjoint intervals from this finite family in decreasing order of length. Every discarded interval meets a selected interval of at least its length, and hence is contained in that interval's threefold dilation. If $I_1, \dots, I_k$ are selected and $S = \sum_i |I_i|$, then $|J| \le 3S$. Let $[a, b]$ be the hull of J and all selected intervals, and put $\ell = b - a$. At most two selected intervals extend beyond J, so

$$S \le \ell \le |J| + S \le 4S, \qquad G := \sum_i |G_i| \ge S/A \ge \lambda \ell.$$

The endpoints $P(a)$ and $P(b)$ are within $76 r_{\max}$ of z and w, respectively. For example, if $a < a_0$, its selected interval contains a_0 and is mapped into a disk of radius $6r$, so $|P(a) - P(a_0)| \le 12r$.

Replace each G_i by its good-scale target path. Replace the complementary subsegments of $[a, b]$ using the prior comparison. At either endpoint p of G_i, both endpoints of I_i are at least $4r$ away. The cross-metric separating-path estimate, applied to I_i, therefore gives a target path surrounding the whole ball $B_{\eta r}(p)$, lying in $B_{r/10}(p)$, of length at most $(d_0/10)K|I_i|$. The endpoints of the two adjacent replacement paths lie in this enclosed ball because $\rho_{\text{old}} < \eta r/4$.

For completeness, the following joining argument does not use a triangle inequality for distances between sets. Initially join each pair of unmatched endpoints by a Euclidean arc in its enclosed ball. Consider one of these possibly nonrectifiable arcs and its separating path. If the chain starts and ends outside the bounded complementary component containing the arc, replace the portion of the chain between its first and last intersections with the separating path by the corresponding portion of that path. If an endpoint of the chain is inside the component, move that endpoint to an intersection of the separating path with the appropriate output ball, and perform the same replacement. Such an intersection exists: the output ball contains the endpoint, is connected, and has radius much larger than the diameter of the separating path, so it cannot lie entirely inside that bounded component. If both endpoints are inside, use the separating path between its intersections with the two output balls. Repeat for each remaining bad joining arc. This removes all such arcs, introduces only rectifiable pieces, and adds at most the sum of the separating-path lengths. Subsequent replacements cannot reintroduce a bad arc.

The resulting endpoints lie in the output balls. In this argument no length is assigned to the auxiliary Euclidean joining arcs: one tracks only the sum of the lengths of the rectifiable pieces, which can increase by at most the length of each inserted separating path. Taking infima over the finitely many nearly minimizing replacement paths, the final total cost is at most

$$C_0(\ell - G) + (1 + d_0)KG + 2d_0K\ell.$$

We have enlarged the actual preliminary errors $d_0/10$ to allow all approximations. If $C_0 \ge K$, use $G \ge \lambda\ell$; if $C_0 < K$, use $G \le \ell$. In both cases the last display is at most $F(C_0, K)\ell \le F(C_0, K)L$, as required. □

Henceforth write $c_* = 513$ for the output constant.

4.2 Polynomial probability of the contraction

Fix a sufficiently large base B, say $B = 4096$. For $0 < s < t < 1$ with $s \le t/4$, let

$$\mathcal{R}_\varepsilon(s, t) = \{B^j : j \in \mathbb{Z},\ \varepsilon^t < B^j < \varepsilon^s\}.$$

Let $\mathcal{S}_\varepsilon$ be any deterministic subset containing at least one third of these radii. The supports $\mathbb{A}_{r/16, 32r}(c)$ are separated at successive radii. More precisely, take the outer scale $64r$, so the

supporting annulus has relative radii $1/1024$ and $1/2$, whereas successive outer scales have ratio at most $1/4096$.

For a fixed center, apply (1.6) to the root good events. The number of available radii is at least

$$\#\mathcal{S}_\varepsilon \geq \frac{t}{4\log B}\log \varepsilon^{-1} - O(1).$$

Choose the exponential rate in (1.6) larger than $16\log B$, and fix p_0 accordingly. A lattice of mesh $\varepsilon^t/100$ covering $B_{R+1}(0)$ has $O_R(\varepsilon^{-2t})$ points. A union bound shows that, with polynomially high probability, each of its points has a good radius in $\mathcal{S}_\varepsilon$. Every point of $B_R(0)$ then satisfies the hypothesis of Lemma 4.1.

Consequently a comparison with constant C and radius $c_*\varepsilon^{t_{\mathrm{old}}}$, where $t_{\mathrm{old}} > t$, improves with polynomially high probability to constant $F(C,K)$ and radius $c_*\varepsilon^s$, where

$$K = \sup_{r\in\mathcal{S}_\varepsilon} \kappa_\varepsilon(r) \quad \text{or} \quad K = \sup_{r\in\mathcal{S}_\varepsilon} \kappa_\varepsilon(r)^{-1}, \tag{4.3}$$

according to the direction. The comparison is simultaneous for all source geodesics contained in $B_R(0)$. The prior comparison event is intersected with the new event. Constants and positive failure exponents can depend on the finitely many bands under consideration. The contraction fraction λ does not.

5 Identification and improvement of the constant

Proposition 5.1. *For every $u > 0$ and $R < \infty$ there exist $s > 0$ and $\beta > 0$ such that with probability at least $1 - O_R(\varepsilon^\beta)$,*

$$\widehat{X}_\varepsilon(B_{c_*\varepsilon^s}(z), B_{c_*\varepsilon^s}(w)) \leq (1+u)D(z,w), \tag{5.1}$$

$$D(B_{c_*\varepsilon^s}(z), B_{c_*\varepsilon^s}(w)) \leq (1+u)\widehat{X}_\varepsilon(z,w), \tag{5.2}$$

whenever, respectively, a D-geodesic or an $\widehat{X}_\varepsilon$-geodesic from z to w is contained in $B_R(0)$.

Proof. Use (2.7)–(2.8) with $\zeta = 1/2$ to obtain an initial constant C_{init}, enlarged if necessary to exceed one, and initial endpoint radius $c_*\varepsilon^{1/2}$. This is valid for every source geodesic in $B_R(0)$ and every subsegment thereof.

Fix $v > 0$ and put $k = (1+v)^{-1} < 1$. Choose d_0 so that

$$T_k := (1 + 3d_0/\lambda)k < 1. \tag{5.3}$$

For the eventual upper comparison also require $(1+3d_0/\lambda)(1+v) < 1+u/2$, choosing v sufficiently small first. Choose N large enough that

$$(1-\lambda)^N C_{\mathrm{init}} < \min\{(1-T_k)/2, u/2\}. \tag{5.4}$$

Take $3N$ disjoint ordered bands with $t_j = 8^{-j}$ and $s_j = t_j/4$, $1 \leq j \leq 3N$. They obey $s_j > t_{j+1}$.

For each band let $n_{\varepsilon,j} = \#\mathcal{R}_\varepsilon(s_j, t_j)$ and $m_{\varepsilon,j} = \lceil n_{\varepsilon,j}/3 \rceil$. Define $q_{\varepsilon,j}$ to be the $m_{\varepsilon,j}$th smallest value of $\kappa_\varepsilon(r)$ over its radii, and $q'_{\varepsilon,j}$ similarly for $\kappa_\varepsilon(r)^{-1}$. These numbers are deterministic.

We claim that for every sufficiently small ε, fewer than N bands have $q_{\varepsilon,j} \leq k$. Otherwise choose a sequence of cutoffs and, by passing to a subsequence, a fixed set of N such bands. In each select $m_{\varepsilon,j}$ radii with $\kappa_\varepsilon(r) \leq k$. Iterate the contraction through those bands. Skipping bands is

harmless since the preceding endpoint radius is still of smaller order than the next smallest root radius. The recursion satisfies

$$C_{n+1} = F(C_n, k), \qquad C_N \le T_k + (1-\lambda)^N C_{\text{init}} < 1. \tag{5.5}$$

For the inequality, note that $T_k \ge k$ is a fixed point of the affine part of $F(\cdot, k)$, and if $C \le T_k$ then $F(C, k) \le T_k$.

This gives, with probability tending to one, a comparison with a fixed constant $c < 1$. Apply it to a D-geodesic realizing $D(O, O')$ for two disjoint rational circles. Such a geodesic exists by compactness of the endpoint circles, lower semicontinuity, and the geodesic property. With probability arbitrarily close to one it is contained in $B_R(0)$ for large deterministic R. We claim that for every deterministic $\rho_\varepsilon \to 0$ and every $c < 1$,

$$\mathbb{P}\Big[\widehat{X}_\varepsilon(B_{\rho_\varepsilon}(O), B_{\rho_\varepsilon}(O')) \le cD(O, O')\Big] \longrightarrow 0. \tag{5.6}$$

For ordinary LFPP this follows from convergence in probability in the lower semicontinuous topology: along an almost surely convergent extracted subsequence, compactness of the endpoint neighborhoods and the moving-endpoint lower bound rule out such an undershoot. For the localized metric, first restrict to $M^{-1} \le D(O, O') \le M$. Confine localized paths of length at most $M + 1$ to a large deterministic disk using Lemma 2.2. On this disk, compact field comparison transfers their lengths to ordinary LFPP by a factor tending to one. Apply the ordinary assertion with any constant strictly between c and one, then let the confinement error tend to zero and $M \to \infty$. This proves (5.6). It contradicts the comparison by $cD(O, O')$ on the event that the source geodesic is contained in $B_R(0)$.

The same argument shows that fewer than N bands have $q'_{\varepsilon,j} \le k$. For this direction use an $\widehat{X}_\varepsilon$-geodesic realizing the distance between the circles. Lemma 2.2 supplies its containment in a large fixed ball with probability arbitrarily close to one, and (2.9) gives convergence of its length. The target distances satisfy

$$D(B_\rho(O), B_\rho(O')) \longrightarrow D(O, O') \quad \text{as } \rho \downarrow 0$$

by compactness and lower semicontinuity. Again a constant below one is impossible.

Thus at least N of the $3N$ bands have both quantiles greater than k. In each of these bands, at most $m_{\varepsilon,j} - 1$ radii satisfy $\kappa_\varepsilon(r) \le k$, and at most $m_{\varepsilon,j} - 1$ satisfy $\kappa_\varepsilon(r) \ge k^{-1}$. At least $n_{\varepsilon,j} - 2(m_{\varepsilon,j} - 1) \ge m_{\varepsilon,j}$ radii therefore satisfy

$$(1+v)^{-1} < \kappa_\varepsilon(r) < 1 + v. \tag{5.7}$$

Iterate the contraction in any N such bands using these radii. Both directions have $K \le 1 + v$, so their final constants are at most

$$(1 + 3d_0/\lambda)(1+v) + (1-\lambda)^N C_{\text{init}} < 1 + u.$$

There are only finitely many choices of bands; take the minimum of their positive probability exponents. Enlarging every final endpoint radius to $c_* \varepsilon^{s_{3N}}$ gives a common $s = s_{3N}$. This proves the proposition. □

6 Interpolation between cutoff values

Lemma 6.1. *For every $a > 0$,*

$$\sup_{\varepsilon \in [(n+1)^{-a}, n^{-a}]} \left| \log \frac{\mathfrak{a}_\varepsilon}{\mathfrak{a}_{n^{-a}}} \right| \longrightarrow 0. \tag{6.1}$$

Proof. Let ε_n be any deterministic choice in the displayed intervals and let $\delta_n = n^{-a}$. For fixed disjoint rational circles, both $X_{\varepsilon_n}(O, O')$ and $X_{\delta_n}(O, O')$ converge in probability to the same positive finite variable $D(O, O')$ by (1.5). Their ratio therefore tends to one.

The ratio of the corresponding *unnormalized* distances also tends to one in probability. Indeed, by Lemma 2.2, nearly minimizing paths at both cutoffs lie in a fixed compact set with probability arbitrarily close to one. On that compact set, (2.2) compares every path length by a factor tending to one almost surely. Comparing the two infima gives the assertion. Dividing the two ratios shows that the deterministic ratio $\mathfrak{a}_{\varepsilon_n}/\mathfrak{a}_{\delta_n}$ tends to one. If (6.1) failed, a deterministic choice of ε_n along a subsequence witnessing failure would contradict this conclusion. □

Lemma 6.2. *Almost surely, for every compact disk K and every $M < \infty$, there exist a finite random R and a positive random cutoff ε_0 such that every ordinary or localized normalized LFPP path of length at most M starting in K is contained in $B_R(0)$ whenever $\varepsilon < \varepsilon_0$.*

Proof. Fix integer disks $K = \overline{B}_k(0)$ and $R > k + 1$. Use (2.8) on $B_{R+1}(0)$ with $\zeta = 1/2$. Choose a sufficiently large that its failure probability is summable on the mesh n^{-a}. Borel–Cantelli, stopping paths on their first hit of $\partial B_R(0)$, and lower semicontinuity give

$$\liminf_{n \to \infty} \widehat{X}_{n^{-a}}(K, \partial B_R(0)) \geq C_0^{-1} D(K, \partial B_R(0)). \tag{6.2}$$

The constant C_0 is independent of k, R. A path from K to $\partial B_R(0)$ can always be stopped at its first hit, so the escape distance can be computed using paths contained in $\overline{B}_R(0)$. Lemmas 2.1 and 6.1 therefore transfer (6.2) to all real cutoffs, both ordinary and localized, without assuming any global confinement at this stage. Take a countable intersection over k, R. By (1.4), choose R with $C_0^{-1} D(K, \partial B_R) > 2M$. The escape distances then exceed M for all sufficiently small cutoffs. This proves the lemma. □

7 Proof of the main theorem

Proof of Theorem 1.1. For each integer $j \geq 1$ and integer $R \geq 1$, apply Proposition 5.1 with $u = 1/j$. Choose a polynomial mesh $n^{-a(j,R)}$ on which the failure probabilities are summable. There is one probability-one event on which all these mesh comparisons hold eventually, all compact field comparisons in Lemma 2.1 hold, and Lemma 6.2 holds. Work on this event. Lemma 6.1 is deterministic and is available for each of the countably many mesh exponents.

We first prove (1.2). Take arbitrary $\varepsilon_n \to 0$ and $(z_n, w_n) \to (z, w)$. If the lower limit of their distances is infinite there is nothing to prove. Otherwise pass to a subsequence on which the distances are bounded and converge to their finite lower limit. The relevant ordinary LFPP geodesics exist and are confined to a random compact set by Lemma 6.2. For a fixed j, choose a large integer R which also contains all bounded-cost localized paths starting near these endpoints; this is possible by the same lemma. Transfer the ordinary geodesic paths to the adjacent cutoff of the mesh for (j, R) using compact field comparison and Lemma 6.1. Their localized mesh lengths

are at most $(1+o(1))X_{\varepsilon_n}(z_n, w_n)$. The localized mesh minimizing geodesics also lie in $B_R(0)$. The reverse comparison (5.2) gives radii $\rho_n \to 0$ such that

$$D(B_{\rho_n}(z_n), B_{\rho_n}(w_n)) \le (1+1/j)(1+o(1))X_{\varepsilon_n}(z_n, w_n).$$

Lower semicontinuity implies that the lower limit of the left side is at least $D(z, w)$. Now let $j \to \infty$. Since the probability-one event did not depend on the endpoint sequences, this proves the required lower-limit assertion simultaneously for all such sequences.

For recovery, fix (z, w) with $D(z, w) < \infty$. The diagonal case $z = w$ is immediate. Otherwise choose a D-geodesic in some $B_R(0)$. For each j, the forward comparison on the mesh for (j, R) provides endpoints approaching z, w whose localized distances are at most $(1+1/j)D(z, w)+o(1)$. Bounded-cost minimizing paths between these endpoints are confined to a fixed larger compact set by Lemma 6.2. Compact field comparison and Lemma 6.1 transfer them to every intervening ordinary cutoff. Thus, for each j, choose $b_j > 0$ so that at every real cutoff $\varepsilon < b_j$ there are endpoints within $1/j$ of z, w whose ordinary normalized distance is at most $(1+2/j)D(z, w)+1/j$. Decrease these thresholds so that $b_j \downarrow 0$. For small ε, let $j(\varepsilon)$ be the largest j such that $\varepsilon < b_j$, and choose the corresponding endpoints. Then $j(\varepsilon) \to \infty$, giving recovery through all cutoffs. No intersection over uncountably many endpoint-dependent probability-one events is taken: these choices are deterministic arguments on the one event fixed at the start of the proof. The already proved lower bound gives the matching lower limit of the recovery values.

If $D(z, w) = \infty$, the lower-limit assertion applied to the constant pair forces $X_\varepsilon(z, w) \to \infty$, so that pair itself is a recovery pair. These are exactly the two defining conditions of the lower semicontinuous topology. The coupling and normalization have remained those of (1.1) throughout. □

A Audit record

This appendix records sequential self-audits, not independent referee reports. It concerns the argument in this manuscript, including the new deterministic contraction, rather than an attempted direct application of a subcritical theorem.

Round 1: source scope and deterministic geometry

The initial draft used an open source annulus whose boundary contained the selected endpoints. It was replaced by $V = \{1/2 \le |z| \le 3\}$. An unnecessary claim that a separating path could be made simple was removed; the joining argument uses its parametrized image directly. The distinction between ordinary and localized rational-circle convergence was repaired by Lemma 2.2. Root-scale units are now explicit, and A is chosen before the error tolerance. The cited annulus concentration, fixed-constant comparison, geodesic existence, and scaling statements were checked in their supercritical scope.

Round 2: locality, dependencies, and constants

The localized field was checked against the precise cutoff definition in [2, Section 2.3]: it is the choice $q = 1$, so Proposition 3.6 applies directly. The source domains, target domains, and convolution dependence neighborhoods all lie inside the root annulus $\mathbb{A}_{r/16,32r}$. Measurability of the uncountable endpoint comparisons is addressed in Remark 3.3; no density of finite-distance points in the metric is assumed. The annulus separation ratio and lattice union bound were checked explicitly. The fraction λ is fixed before the error tolerance, and every iteration uses only finitely many exponent

bands. The output-ball constant was made uniformly 513 to remove a strict-boundary ambiguity. The dependency chain is fixed-constant comparison, confinement in probability, localized circle convergence, contraction, near-one comparison, normalizer interpolation, and finally almost sure confinement. In particular, the normalizer argument does not use almost sure metric convergence.

Round 3: exact conclusion and source recheck

The final pass checked the proof backwards from the two defining conditions of the lower semi-continuous topology. Bounded-cost paths are confined before any compact field interpolation is used. The escape proof stops paths at their first exit and therefore does not assume the confinement it proves. The final probability-one event involves only countably many mesh exponents and Euclidean disks; the endpoint sequences are arbitrary after that event is fixed. Recovery through every real cutoff is now given by an explicit diagonal choice. Infinite distances are handled by the lower-limit condition, with no subtraction of infinities. The normalizer is the same unit-square crossing median at every step. The fixed-annulus LFPP tightness input was checked directly in [6, Proposition 4.1], and the field bound was attributed directly to [2, Lemma 2.3]. A locality overstatement in the separating-path lemma was corrected: its event is asserted only for the truncated metric $Y_{\delta,L}$, with the dependence-radius condition stated explicitly. Ordinary LFPP is used there only to bound marginal probabilities. This pass also made explicit why the closed-domain path lengths remain local and why the joining argument never assigns a finite metric cost to an arbitrary Euclidean arc. No unresolved step was identified in these three self-audits; they are not a substitute for independent mathematical review.

References

[1] J. Ding and E. Gwynne. *Uniqueness of the critical and supercritical Liouville quantum gravity metrics*. arXiv:2110.00177, version 3. Theorems 1.3 and 1.8, Definition 1.7.

[2] J. Ding and E. Gwynne. *Up-to-constants comparison of Liouville first passage percolation and Liouville quantum gravity*. arXiv:2108.12060, version 1. Lemmas 2.2, 3.1 and 3.7, Proposition 3.6, Theorem 1.9.

[3] J. Ding and E. Gwynne. *Regularity and confluence of geodesics for the supercritical Liouville quantum gravity metric*. arXiv:2104.06502, version 4. Lemma 2.12.

[4] J. Pfeffer. *Weak Liouville quantum gravity metrics with matter central charge* $\mathbf{c} \in (-\infty, 25)$. arXiv:2104.04020. Theorem 1.7, Propositions 1.10 and 1.12, Lemma 3.12.

[5] C. Devlin VI. *Almost sure convergence of Liouville first passage percolation*. arXiv:2309.08001, version 2. Lemma 4.2 is cited for comparison with the field interpolation estimate; no subcritical metric estimate is imported.

[6] J. Ding and E. Gwynne. *Tightness of supercritical Liouville first passage percolation*. arXiv:2005.13576. Proposition 4.1.